$$
\begin{aligned}
&C_{\phi D}^{(11)}\ {}^{(22)} \to \frac{1}{16\,\pi^2}\ \Big(\frac{1}{9}\ g_2{}^4\, LF_{3,0}[m_2] + \frac{1}{6}\ g_2{}^4\, LF_{4,-1}[m_2] - \frac{8}{45}\ g_2{}^4\, LF_{5,-2}[m_2] - \\
&\frac{1}{27}\sum_p g_1{}^4\, LF_{3,0}\big[m_{\tilde{d}}{}^p\big] + \frac{5}{72}\sum_p g_1{}^4\, LF_{4,-1}\big[m_{\tilde{d}}{}^p\big] - \frac{4}{135}\sum_p g_1{}^4\, LF_{5,-2}\big[m_{\tilde{d}}{}^p\big] - \frac{1}{9}\sum_p g_1{}^4\, LF_{3,0}\big[m_{\tilde{e}}{}^p\big] + \\
&\frac{5}{24}\sum_p g_1{}^4\, LF_{4,-1}\big[m_{\tilde{e}}{}^p\big] - \frac{4}{45}\sum_p g_1{}^4\, LF_{5,-2}\big[m_{\tilde{e}}{}^p\big] + \frac{1}{18}\sum_p \left(-g_1{}^4 + g_2{}^4\right) LF_{3,0}\big[m_{\tilde{l}}{}^p\big] + \\
&\frac{5}{48}\sum_p \left(g_1{}^4 - g_2{}^4\right) LF_{4,-1}\big[m_{\tilde{l}}{}^p\big] - \frac{2}{45}\sum_p \left(g_1{}^4 - g_2{}^4\right) LF_{5,-2}\big[m_{\tilde{l}}{}^p\big] - \frac{1}{54}\sum_p \left(g_1{}^4 - 9\ g_2{}^4\right) LF_{3,0}\big[m_{\tilde{q}}{}^p\big] + \\
&\frac{5}{144}\sum_p \left(g_1{}^4 - 9\ g_2{}^4\right) LF_{4,-1}\big[m_{\tilde{q}}{}^p\big] - \frac{2}{135}\sum_p \left(g_1{}^4 - 9\ g_2{}^4\right) LF_{5,-2}\big[m_{\tilde{q}}{}^p\big] - \\
&\frac{4}{27}\sum_p g_1{}^4\, LF_{3,0}\big[m_{\tilde{u}}{}^p\big] + \frac{5}{18}\sum_p g_1{}^4\, LF_{4,-1}\big[m_{\tilde{u}}{}^p\big] - \frac{16}{135}\sum_p g_1{}^4\, LF_{5,-2}\big[m_{\tilde{u}}{}^p\big] + \\
&\frac{1}{18}\left(-g_1{}^4 + g_2{}^4\right) LF_{3,0}[\tilde{\mu}] + \frac{1}{12}\left(-g_1{}^4 + g_2{}^4\right) LF_{4,-1}[\tilde{\mu}] + \frac{4}{45}\left(g_1{}^4 - g_2{}^4\right) LF_{5,-2}[\tilde{\mu}] + \\
&\frac{1}{16}\ g_1{}^4\, LF_{2,2,-1}[m_1,\ \tilde{\mu}] + \frac{1}{8}\ g_1{}^4\, m_1{}^2\, LF_{2,2,0}[m_1,\ \tilde{\mu}] - \frac{2}{3}\ g_2{}^4\, LF_{2,1,0}[m_2,\ \tilde{\mu}] - \\
&\frac{17}{48}\ g_2{}^4\, LF_{2,2,-1}[m_2,\ \tilde{\mu}] + \frac{1}{8}\ g_2{}^4\, m_2{}^2\, LF_{2,2,0}[m_2,\ \tilde{\mu}] + \frac{1}{3}\ g_2{}^4\, LF_{3,1,-1}[m_2,\ \tilde{\mu}] + \\
&\frac{1}{3}\ g_2{}^4\, LF_{3,2,-2}[m_2,\ \tilde{\mu}] - \frac{1}{6}\ g_1{}^2\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) LF_{2,2,0}\big[m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{1}{12}\ g_1{}^2\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) LF_{3,2,-1}\big[m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p\big] - \frac{1}{6}\ g_1{}^2\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) \\
&\quad LF_{2,2,0}\big[m_{\tilde{e}}{}^r,\ m_{\tilde{l}}{}^p\big] + \frac{1}{12}\ g_1{}^2\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) LF_{3,2,-1}\big[m_{\tilde{e}}{}^r,\ m_{\tilde{l}}{}^p\big] + \\
&\frac{1}{6}\ g_1{}^2\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) LF_{3,1,0}\big[m_{\tilde{l}}{}^p,\ m_{\tilde{e}}{}^r\big] + \frac{1}{6}\ g_1{}^2\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) \\
&\quad LF_{3,2,-1}\big[m_{\tilde{l}}{}^p,\ m_{\tilde{e}}{}^r\big] - \frac{1}{8}\left(3\ g_1{}^2 - g_2{}^2\right)\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) LF_{4,1,-1}\big[m_{\tilde{l}}{}^p,\ m_{\tilde{e}}{}^r\big] + \\
&\frac{1}{6}\left(g_1{}^2 - g_2{}^2\right)\left(\overline{a_e}^{pr}\, a_e{}^{pr} + \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e{}^{pr}\right) LF_{5,1,-2}\big[m_{\tilde{l}}{}^p,\ m_{\tilde{e}}{}^r\big] + \frac{1}{6}\ g_1{}^2\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) \\
&\quad LF_{3,1,0}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{d}}{}^r\big] + \frac{1}{6}\ g_1{}^2\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) LF_{3,2,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{d}}{}^r\big] - \\
&\frac{1}{8}\left(5\ g_1{}^2 - 3\ g_2{}^2\right)\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) LF_{4,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{d}}{}^r\big] + \\
&\frac{1}{2}\left(g_1{}^2 - g_2{}^2\right)\left(\overline{a_d}^{pr}\, a_d{}^{pr} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\right) LF_{5,1,-2}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{d}}{}^r\big] + \\
&\frac{1}{12}\left(g_1{}^2 + 3\ g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{2,2,0}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r\big] - \\
&\frac{1}{24}\left(g_1{}^2 + 3\ g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{3,2,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r\big] - \\
&\frac{1}{12}\left(g_1{}^2 + 3\ g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{3,1,0}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^p\big] - \\
&\frac{1}{12}\left(g_1{}^2 + 3\ g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{3,2,-1}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^p\big] - \\
&\frac{1}{4}\left(g_1{}^2 - 3\ g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{4,1,-1}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{1}{2}\left(g_1{}^2 - g_2{}^2\right)\left(\overline{a_u}^{pr}\, a_u{}^{pr} + \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\right) LF_{5,1,-2}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{3}{2}\ \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt}\, LF_{2,1,0}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r\big] - \frac{3}{2}\ \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt}\, LF_{3,1,-1}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r\big] + \\
&\frac{1}{8}\ g_1{}^4\, LF_{3,2,-2}[\tilde{\mu},\ m_1] - \frac{1}{8}\ g_1{}^4\, m_1{}^2\, LF_{3,2,-1}[\tilde{\mu},\ m_1] + \frac{5}{12}\left(g_1{}^4 - g_1{}^2\, g_2{}^2\right) LF_{4,1,-2}[\tilde{\mu},\ m_1] - \\
&\frac{1}{8}\ g_1{}^4\, LF_{4,2,-3}[\tilde{\mu},\ m_1] + \frac{1}{8}\ g_1{}^4\, m_1{}^2\, LF_{4,2,-2}[\tilde{\mu},\ m_1] + \frac{1}{3}\left(-g_1{}^4 + g_1{}^2\, g_2{}^2\right) LF_{5,1,-3}[\tilde{\mu},\ m_1] + \\
&\frac{4}{3}\ g_2{}^4\, LF_{3,1,-1}[\tilde{\mu},\ m_2] + \frac{31}{24}\ g_2{}^4\, LF_{3,2,-2}[\tilde{\mu},\ m_2] - \frac{1}{8}\ g_2{}^4\, m_2{}^2\, LF_{3,2,-1}[\tilde{\mu},\ m_2] + \\
&\frac{1}{4}\ g_2{}^2\left(5\ g_1{}^2 - 9\ g_2{}^2\right) LF_{4,1,-2}[\tilde{\mu},\ m_2] - \frac{5}{8}\ g_2{}^4\, LF_{4,2,-3}[\tilde{\mu},\ m_2] + \frac{1}{8}\ g_2{}^4\left(m_2{}^2 + 4\ \tilde{\mu}^2\right) LF_{4,2,-2}[\tilde{\mu},\ m_2] + \\
&\left(-g_1{}^2\, g_2{}^2 + g_2{}^4\right) LF_{5,1,-3}[\tilde{\mu},\ m_2] + 3\ g_1{}^2\, g_2{}^2\, \tilde{\mu}^2\, LF_{2,2,1,-1}[m_2,\ \tilde{\mu},\ m_1] - \\
&2\ g_1{}^2\, g_2{}^2\, \tilde{\mu}^2\, LF_{3,2,1,-2}[m_2,\ \tilde{\mu},\ m_1] - \frac{3}{4}\ \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,2,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t\big] + \\
&\frac{3}{4}\ y_d{}^{sr}\, \overline{y_u}^{st}\left(-\overline{a_d}^{pr}\, a_u{}^{pt} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_u{}^{pt}\right) LF_{2,2,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt}\, LF_{2,2,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r\big] + \\
&\frac{3}{2}\ \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,1,1,0}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^p\big] - \frac{3}{2}\ \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{3,1,1,-1}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{3}{2}\ y_d{}^{sr}\, \overline{y_u}^{st}\left(\overline{a_d}^{pr}\, a_u{}^{pt} - \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_u{}^{pt}\right) LF_{2,1,1,0}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{3}{2}\ y_d{}^{sr}\, \overline{y_u}^{st}\left(-\overline{a_d}^{pr}\, a_u{}^{pt} + \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_u{}^{pt}\right) LF_{3,1,1,-1}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p\big] - \\
&\frac{3}{2}\ \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt}\, LF_{2,1,1,0}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^s\big] + \\
&\frac{3}{2}\ \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt}\, LF_{3,1,1,-1}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^s\big] - \frac{1}{8}\ g_1{}^2\, g_2{}^2\, LF_{2,1,1,-1}[\tilde{\mu},\ m_1,\ m_2] - \\
&\frac{1}{4}\ m_1\, m_2\, g_1{}^2\, g_2{}^2\, LF_{2,1,1,0}[\tilde{\mu},\ m_1,\ m_2] - \frac{1}{4}\ g_1{}^2\, g_2{}^2\, LF_{3,1,1,-2}[\tilde{\mu},\ m_1,\ m_2] + \\
&\frac{1}{4}\ g_1{}^2\, g_2{}^2\left(m_1\, m_2 + 4\ \tilde{\mu}^2\right) LF_{3,1,1,-1}[\tilde{\mu},\ m_1,\ m_2] + \frac{1}{4}\ g_1{}^2\, g_2{}^2\, LF_{4,1,1,-3}[\tilde{\mu},\ m_1,\ m_2] - \\
&\frac{1}{4}\ g_1{}^2\, g_2{}^2\left(m_1\, m_2 + 2\ \tilde{\mu}^2\right) LF_{4,1,1,-2}[\tilde{\mu},\ m_1,\ m_2] - 2\ g_1{}^2\, g_2{}^2\, \tilde{\mu}^2\, LF_{3,2,1,-2}[\tilde{\mu},\ m_2,\ m_1] - \\
&\frac{1}{4}\ \tilde{\mu}^2\, \overline{y_e}^{st}\, y_e{}^{sr}\, \overline{a_e}^{pr}\, a_e{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{l}}{}^p,\ m_{\tilde{l}}{}^s,\ m_{\tilde{e}}{}^r,\ m_{\tilde{e}}{}^t\big] - \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_d{}^{sr}\, \overline{a_d}^{pr}\, a_d{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s,\ m_{\tilde{d}}{}^r,\ m_{\tilde{d}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, y_d{}^{sr}\, \overline{y_u}^{st}\, \overline{a_d}^{pr}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s,\ m_{\tilde{d}}{}^r,\ m_{\tilde{u}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s,\ m_{\tilde{d}}{}^t,\ m_{\tilde{u}}{}^r\big] - \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^r,\ m_{\tilde{u}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^r,\ m_{\tilde{u}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^s\big] + \\
&\frac{3}{8}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^t\big] - \\
&\frac{3}{4}\left(\tilde{\mu}^4\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt} + \overline{a_d}^{pr}\, a_u{}^{pt}\left(a_d{}^{sr}\, \overline{a_u}^{st} - \tilde{\mu}^2\, y_d{}^{sr}\, \overline{y_u}^{st}\right)\right) LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^s\big] + \\
&\frac{3}{8}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^t,\ m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^r\big] - \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r,\ m_{\tilde{u}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r,\ m_{\tilde{u}}{}^t\big] + \\
&\frac{3}{4}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{3}{8}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^r,\ m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^t\big] - \\
&\frac{3}{4}\left(\tilde{\mu}^4\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt} + \overline{a_d}^{pr}\, a_u{}^{pt}\left(a_d{}^{sr}\, \overline{a_u}^{st} - \tilde{\mu}^2\, y_d{}^{sr}\, \overline{y_u}^{st}\right)\right) LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p\big] + \\
&\frac{3}{8}\ \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u{}^{sr}\, \overline{a_u}^{pr}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^s,\ m_{\tilde{u}}{}^t,\ m_{\tilde{q}}{}^p,\ m_{\tilde{u}}{}^r\big] - \\
&\frac{3}{2}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{2,1,1,1,0}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s\big] + \\
&\frac{3}{2}\ \tilde{\mu}^2\, \overline{y_d}^{st}\, y_u{}^{sr}\, a_d{}^{pt}\, \overline{a_u}^{pr}\, LF_{3,1,1,1,-1}\big[m_{\tilde{u}}{}^r,\ m_{\tilde{d}}{}^t,\ m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s\big] + \\
&\frac{3}{2}\left(\tilde{\mu}^4\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt} + \overline{a_d}^{pr}\, a_u{}^{pt}\left(a_d{}^{sr}\, \overline{a_u}^{st} - \tilde{\mu}^2\, y_d{}^{sr}\, \overline{y_u}^{st}\right)\right) LF_{2,1,1,1,0}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s\big] - \\
&\frac{3}{2}\left(\tilde{\mu}^4\, \overline{y_d}^{pr}\, y_d{}^{sr}\, \overline{y_u}^{st}\, y_u{}^{pt} + \overline{a_d}^{pr}\, a_u{}^{pt}\left(a_d{}^{sr}\, \overline{a_u}^{st} - \tilde{\mu}^2\, y_d{}^{sr}\, \overline{y_u}^{st}\right)\right) LF_{3,1,1,1,-1}\big[m_{\tilde{u}}{}^t,\ m_{\tilde{d}}{}^r,\ m_{\tilde{q}}{}^p,\ m_{\tilde{q}}{}^s\big]\Big)
\end{aligned}
$$